

AI Features — Frontend Testing Guide

Complete End-to-End Testing Flow for All Integrated AI APIs

This guide documents the complete, feature-by-feature testing flow for every AI API integrated into the platform — Lead Summarization, AI Feedback, Prompt Versioning, Token Usage Tracking, Automation Rules & Runs, and the Knowledge Base (RAG) pipeline. Follow each flow in order to verify the frontend correctly triggers and reflects each backend endpoint.

Step 1 — Start the Frontend Dev Server

Before testing any AI feature, ensure the frontend development server is running:

```
cd portfolio
npm run dev
```

Then open your browser at `http://localhost:5173` (or whichever port the terminal displays).

Step 2 — Testing Flow, Feature by Feature

■ Flow 1: AI Lead Summarization & AI Feedback Submission

Endpoints Tested:

`POST /api/v1/ai/summarize-lead` `POST /api/v1/ai/feedback`

- 1 Go to **Sidebar** → **Leads / Leads List**.
- 2 Click any lead to open its **Detail View** (`/leads/:id`).
- 3 Below the internal agent notes, locate the **“AI Lead Intelligence & Summarization”** card.
- 4 Click **“Summarize with AI”** — this sends a request to `/api/v1/ai/summarize-lead`. The response renders an **AI Summary** and **Detected Intent** card (e.g. *High Purchase Intent — Pricing Inquiry*).
- 5 In the summary card footer, click **Thumbs Up / Thumbs Down** or **“Rate & Comment”**.
- 6 Select a star rating (1–5), enter a feedback comment, and click **Submit Feedback** — this logs the request to `/api/v1/ai/feedback`.

■ Flow 2: Automation Rules Management

Endpoints Tested:

`POST /api/v1/ai/automation-rules` `GET /api/v1/ai/automation-rules/{organization_id}` `POST /api/v1/ai/automation-runs`

- 1 Go to **Sidebar** → **Config/Settings** → **AI Automations** (`/ai-automations`).
- 2 Select the **“Automation Rules”** tab — all active rules render here from the backend.
- 3 Click **“+ New Automation Rule”** (top right) to open the creation modal.
- 4 Fill in Rule Name, Trigger Event (e.g. *Lead Created*), Action Type (e.g. *Send AI WhatsApp*), and Conditions JSON, then click **Create Rule** — this calls `POST /api/v1/ai/automation-rules`.
- 5 In the rules table, click **“Test Run”** next to any rule — this triggers `POST /api/v1/ai/automation-runs` and logs the execution details.

■ Flow 3: Execution History & Payload Inspector

Endpoints Tested:

`Combined AutomationRun & AIRun History`

- 1 On the AI Automations page, click the **“Execution & AI Runs”** tab (or go to **Sidebar** → **AI Runs**, `/ai-runs`).
- 2 Review the list of recent AI executions and their status (*success*, *failed*).
- 3 Click **“Inspect Payload”** / **“View JSON Payload”** on any entry to open a code-formatted dark JSON modal showing the exact `input_data`, `output_data`, and `execution_duration_ms`.

■ Flow 4: Prompt Version Control

Endpoints Tested:

`POST /api/v1/ai/prompts` `GET /api/v1/ai/prompts/active/{feature_name}`

- 1 On the AI Automations page, select the **“Prompt Versioning”** tab.
- 2 Review the list of active and inactive prompt versions (e.g. *lead_summary*, *copilot_chat*).
- 3 Click **“+ Create Prompt Version”**, fill in the system prompt text, version number, and feature name, then submit — this calls `POST /api/v1/ai/prompts`.
- 4 Toggle the **“Active Version”** switch on any version to change the active prompt.

■ Flow 5: Knowledge Base & Vector Chunks (RAG)

Endpoints Tested:

`POST /api/v1/ai/knowledge/documents` `POST /api/v1/ai/knowledge/chunks`

- 1 On the AI Automations page, select the **“Knowledge Base (RAG)”** tab.

- 2 Click “+ **Register Document**”, enter Document Title, File URL (<https://...>), and Source Type (PDF, DOCX, TXT) — this calls `POST /api/v1/ai/knowledge/documents`.
- 3 On the registered document, click “+ **Add Chunk to Doc**” (or top-level “+ **Store Vector Chunk**”), fill in a text snippet and its embedding vector (e.g. `[0.012, -0.045, 0.881]`), then click **Store Chunk** — this calls `POST /api/v1/ai/knowledge/chunks`.

■ Flow 6: Token Usage & Cost Metrics

Endpoints Tested:

`POST /api/v1/ai/usage`

- 1 On the AI Automations page, select the “**Token & Cost Metrics**” tab.
- 2 Review the top KPI cards for total tokens consumed and calculated dollar cost.
- 3 Click “+ **Record Usage Entry**” to log a custom token usage entry — this calls `POST /api/v1/ai/usage`.

Summary Checklist

Feature / API	Main UI Location	Trigger Action
Lead Summarize	/leads/:id	"Summarize with AI" button
AI Feedback	Lead Detail or AI Feedback Tab	Rating stars & Comment modal
Automation Rules	/ai-automations → Rules Tab	"+ New Automation Rule" / "Test Run"
Execution Runs	/ai-runs	"Inspect Payload" modal
Prompt Versions	/ai-automations → Prompts Tab	"+ Create Prompt Version" / Active toggle
RAG Knowledge Base	/ai-automations → Knowledge Tab	"+ Register Document" & "+ Store Chunk"
Token Usage	/ai-automations → Usage Tab	"+ Record Usage Entry"

Base URL for all UI routes above: <http://localhost:5173>